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# FEEDBACK OSCILLATORS

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This **Basic Electronics Tutorials eBook** is focused on the different types of electronic feedback oscillator with the information presented within this ebook provided “as-is” for general information purposes only.

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## 1. FUNDAMENTALS OF AN OSCILLATOR

In its broadest terms, **Electronic Feedback Oscillators** are wave generating circuits, or devices, which have the ability to generate their own repetitive output signal. They are generally classified according to the type and shape of the output signal they produce, as well as the electronic circuitry required to produce the desired oscillations.

The shape of the periodic output waveform can be purely sinusoidal, or non-sinusoidal. Oscillators can have a wide range of different wave shapes and frequencies, or just one single shape at a preset frequency. For example, a 60Hz sinusoidal waveform. Note that non-sinusoidal oscillators are known commonly as relaxation oscillators.

If negative feedback is applied to an amplifier, the amplifiers gain is decreased but its stability is increased. With positive feedback, the gain is increased but the stability is decreased. Thus the amplifier has now become an oscillator giving an alternating output.

Then an oscillator is nothing more than a self-sustaining amplifier which uses positive regenerative (in-phase) feedback to create a periodic output signal without the need for any externally applied input signal. However, the feedback signal should have enough magnitude to ensure that maintained oscillations take place.

*Oscillators are unstable amplifiers with positive feedback and no input*

One of the many problems of an electronic circuit designer is stopping amplifiers from oscillating while trying to get oscillators to oscillate. Then we can correctly say that an electronic oscillator is an unstable amplifier.

All oscillators convert DC power from the power supply into some form of periodic output wave shape with computers, clocks, watches, radios, and test equipment among the many devices that use electronic oscillators.

They can produce either a continuous sinusoidal waveform, square, sawtooth or triangular shaped waveforms or just a train of repetitive pulses of a variable or constant width. But the main requirement of any type of oscillator is that its output is uniform and stable at the desired frequency. That is, oscillator stability.

So for any electronic circuit to operate as an oscillator, it must contain the following three major characteristics:

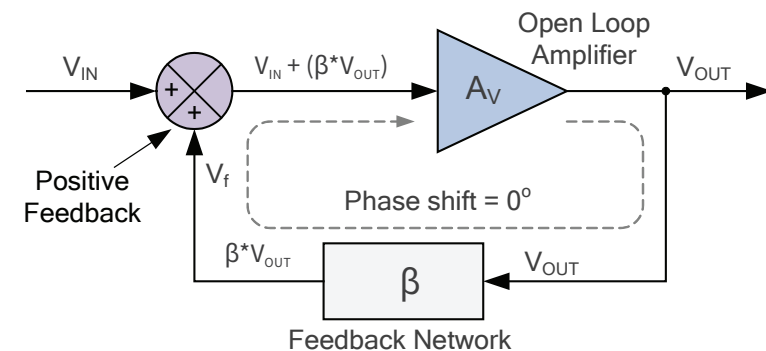
- Some form of Voltage Amplification
- Sufficient Positive Feedback (regeneration)
- Some Form of Frequency Determining Feedback Network

As stated previously, a feedback oscillator is basically an amplifier with positive feedback from its output to its input enabling it to sustain the output oscillations without the need for any external input signal. Any small signal feedback amplifier such as an operational amplifier, FET or bipolar transistor amplifier with an open-loop gain equal to or slightly greater than one (unity) could be used as an oscillator circuit.

Thus unlike an amplifier circuit, there is no external AC input signal required to cause the oscillator to work, as the DC supply energy is converted by the oscillator into AC energy at the required frequency. Thus, oscillators are essentially closed loop regenerative circuits which use DC power to maintain AC oscillations.

Figure 1. Shows a basic principles of an oscillator which has an amplifier with an open-loop voltage gain of  $A_V$  and having a feedback fraction or ratio of  $\beta$  (beta).

**FIGURE 1. BASIC OSCILLATOR FEEDBACK CIRCUIT**



Since feedback is the process by which a portion of the output signal is returned back to the input, being an important characteristic of any oscillator circuit. We can derive the generalised equation for Figure 1. for a basic oscillator circuit as being:

$$\text{Open-loop Voltage Gain, } A_V = \frac{V_{OUT}}{V_{IN}}$$

$$V_{OUT} = A_V(V_{IN} + V_f)$$

$$V_f = \beta V_{OUT} = A_V \beta V_{OUT}$$

$$V_{OUT} = A_V(V_{IN} + \beta V_{OUT})$$

$$\therefore A_V V_{IN} = V_{OUT}(1 - A_V \beta)$$

Reorganising a little, we get the oscillator gain with feedback as being:

$$A_f = \frac{V_{OUT}}{V_{IN}} = \frac{A_V}{1 - A_V \beta}$$

Where:  $A_f$  is the closed-loop gain,  $-A_V \beta$  (or simply  $-A\beta$ ) is the loop gain.

Since we have no input signal,  $V_f$  effectively becomes equal to  $V_{IN}$  therefore, the required condition of  $A\beta = 1$  is satisfied so the circuit maintains self-sustained oscillations. The requirement of  $A\beta = 1$  (giving  $360^\circ$  phase shift) for oscillations is commonly called the **Barkhausen Criterion** where the amplifier gain reduces by a factor of:  $(1 - A\beta)$ .

As we will see later, the feedback fraction or ratio,  $\beta$  is a function of frequency. Thus the *Barkhausen criterion* will only be satisfied for one predetermined frequency of the output waveform. Then for sustained output oscillations to occur:

1. The phase shift of the loop-gain through the amplifier and feedback circuit is equivalent to  $0^\circ$  or  $360^\circ$  ( $A\beta V_{IN}$  is in phase with  $V_{IN}$ )
2. Oscillations will be sustained at the oscillator frequency if the magnitude of loop gain is equal to 1 (unity)

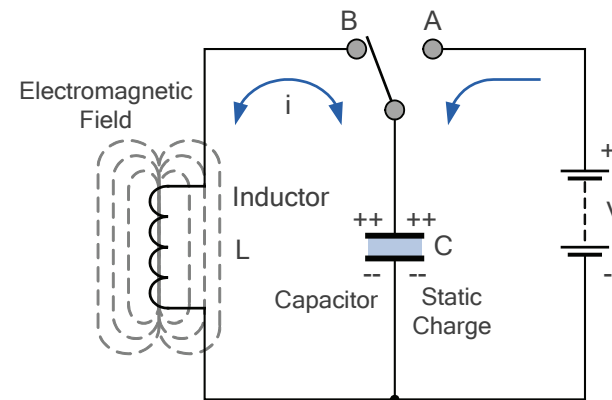
Note that if the  $A\beta$  loop gain is less than 1, the output oscillations will reduce each and every cycle until they eventually die out. If the  $A\beta$  loop gain is greater than 1, output oscillations build-up until eventually the output saturates at the supply voltage level.

## 2. LC OSCILLATORS

LC feedback, also called harmonic oscillators, work because they overcome the losses of their feedback circuit with a portion of the output power being returned to the input circuit. This type of oscillator commonly employs a tuned **LC** (Inductor Capacitor) circuit. When an inductor and a capacitor are connected together in parallel across a supply, they form what is commonly called a **tank circuit**.

When an LC tank circuit is “excited” by an external DC power source, it wants to oscillate. LC oscillators use resonant tank circuits to store energy alternately between an inductor and a capacitor with the frequency of oscillations determined by the resonant condition of the reactive pair. An LC oscillator can generate a purely sinusoidal output signal of constant amplitude and frequency by the interaction of their LC tank circuit.

**FIGURE 2. BASIC LC TANK CIRCUIT**



The circuit of Figure 2. consists of an inductive coil, **L** and capacitor, **C**. The capacitor stores energy in the form of an *electrostatic field* producing a static voltage across its plates, while the inductive coil stores its energy in the form of an *electromagnetic field*.

The capacitor charges up to the DC supply voltage, **V** when the switch is in position **A**. When the capacitor is fully charged the switch changes to position **B**. The charged capacitor is now connected in parallel across the inductive coil so the capacitor begins to discharge itself through the coil. The voltage across **C** starts falling as the current through the coil begins to rise.

When the capacitor is completely discharged the energy that was originally stored by the capacitor as an electrostatic field is now stored in the inductive coil as an electromagnetic field around the coils windings. Since there is no external voltage to maintain the current within the coil, it starts to fall as the electromagnetic field begins to collapse producing a back emf.

This current charges up capacitor with the opposite polarity to its original charge. C continues to charge up until the current reduces to zero and the electromagnetic field of the coil has collapsed completely.

*The resonance of a tank circuit occurs when the inductive reactance equals the capacitive reactance*

The energy originally introduced into the circuit through the switch, has been returned to the capacitor which again has an electrostatic voltage potential across it, although it is now of the opposite polarity. The capacitor now starts to discharge again back through the coil and the whole process is repeated.

Thus a circulating current flows inside the tank producing a back and forth oscillatory motion. This action would continue indefinitely if it was not for energy losses within the tank circuit. Then to keep the oscillations going in an LC tank circuit, we have to replace all the energy lost in each oscillation while maintaining the amplitude of these oscillations at a constant level. The amount of energy replaced must therefore be equal to the energy lost during each cycle.

The frequency of the oscillatory voltage depends upon the value of the inductance and capacitance in the LC tank circuit. *Inductive reactance*,  $X_L$  increases with increasing frequency, while *capacitive reactance*,  $X_C$  decreases with increasing frequency.

For resonance to occur in the tank circuit, there must be one frequency point where the value of  $X_L$  is the exactly same as the value of  $X_C$ . That is  $X_L = X_C$  thus cancelling each other out. The frequency value at which this happens is given as:

$$X_L = 2\pi fL \quad \text{and} \quad X_C = \frac{1}{2\pi fC}$$

So, at the resonant frequency:  $X_L = X_C$

$$\therefore 2\pi fL = \frac{1}{2\pi fC}$$

$$2\pi f^2L = \frac{1}{2\pi C}$$

$$\therefore f^2 = \frac{1}{(2\pi)^2 LC}$$

$$f = \frac{\sqrt{1}}{\sqrt{(2\pi)^2 LC}}$$

Then by simplifying the above equation we get the final equation for *Resonant Frequency*,  $f_r$  of a tuned LC circuit as being:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Where:  $f_r$  is the oscillation frequency in Hertz (Hz),  $L$  is the Inductance of the coil winding in Henries (H), and  $C$  is the capacitance value of the capacitor in Farads (F).

Note that the above equation shows that if either  $L$  or  $C$  are decreased in value, the oscillating output frequency would increase. Feedback oscillator output frequency is commonly given the abbreviation of ( $f_r$ ) to identify it as the “resonant frequency”.

LC sinusoidal feedback oscillators are capable of generating high frequency sine waves for use in radio frequency (RF) type applications using Bipolar Transistors or Field Effect Transistor's.

*LC oscillators use a parallel inductive-capacitive circuit to generate output oscillations*

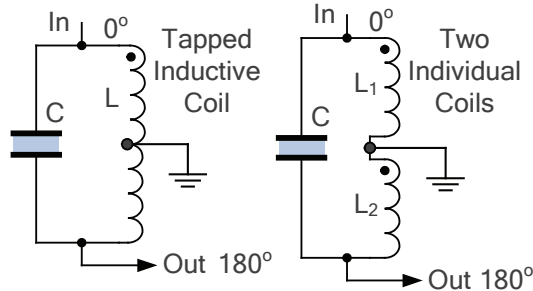
Sinusoidal oscillators come in many different forms being named after their inventors with the most common being the *Hartley LC Oscillator*, *Colpitts LC Oscillator*, *Armstrong Oscillator* and *Clapp Oscillator* to name a few.

### 3. THE HARTLEY OSCILLATOR

The **Hartley Oscillator** design uses two inductive coils connected in series with a parallel capacitor across them to form its resonance tank circuit. That is feedback is achieved by way of an inductive divider to produce the required sinusoidal oscillations.

The tuned LC tank circuit can be made from any configuration that uses either a single centre or offset tapped coil, or a pair of series connected coils which are themselves in parallel with a variable capacitor,  $C$  as shown in Figure 3.

FIGURE 3. SHUNT-FED HARTLEY OSCILLATOR TANK CIRCUIT



Then a tapped inductor in the oscillators tank circuit clearly identifies it as a Hartley oscillator configuration.

The Hartley oscillator is often referred to as a split-inductance oscillator, because coil  $L$  is tapped along its length. In effect, inductance  $L$  acts like two separate coils in very close proximity, with or without inductive coupling.

Then as we can see in Figure 3, the output voltage is developed across one of these two inductors. The output voltage appears across  $L_2$  and the feedback voltage appearing across inductor  $L_1$ . Then the feedback fraction,  $\beta$  is given as:  $\beta = L_2/L_1$ . Note that for oscillations to start, the amplifiers voltage gain must be greater than  $1/\beta$ .

If the equivalent inductance  $L_{EQ}$  is defined as:  $L_{EQ} = L_1 + L_2$ . The frequency of oscillations for the Harley oscillator is given as:

$$f_r = \frac{1}{2\pi\sqrt{CL_{EQ}}} = \frac{1}{2\pi\sqrt{C(L_1+L_2)}}$$

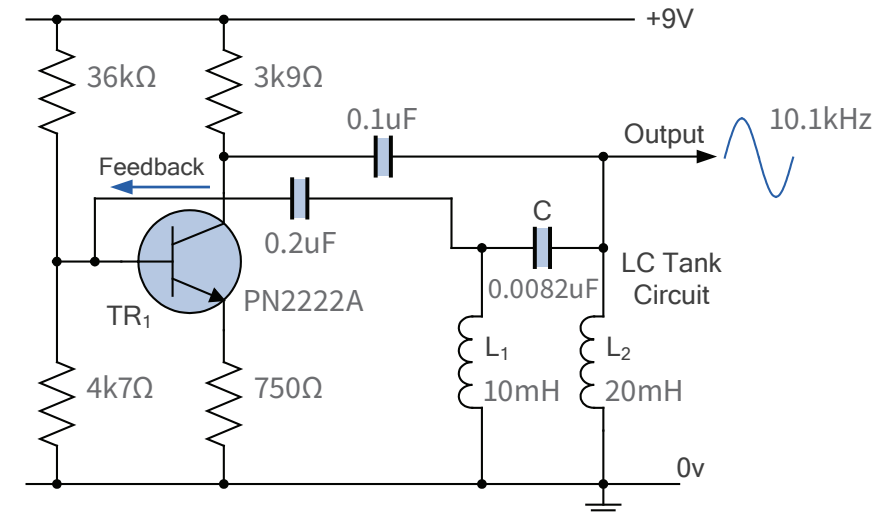
Note that this basic formula is adequate when there is no mutual inductance ( $M$ ) between  $L_1$  and  $L_2$ , but requires modification to:  $L_{EQ} = L_1 + L_2 + 2M$  if their mutual inductance is considerable.

So for example, the frequency of oscillations of a Hartley oscillator with one inductor of  $10\text{mH}$  and another of  $20\text{mH}$  in parallel with a  $0.0082\mu\text{F}$  capacitor would be:

$$\frac{1}{2\pi\sqrt{C(L_1+L_2)}} = \frac{1}{2\pi\sqrt{0.0082 \times 10^{-6}(0.01+0.02)}} = 10146 \text{ Hz or } 10.1\text{kHz}$$

The  $10\text{kHz}$  shunt-fed Hartley oscillator with the bipolar transistor connected in its common emitter connection is shown in Figure 4.

FIGURE 4. SHUNT-FED HARTLEY OSCILLATOR BJT CIRCUIT



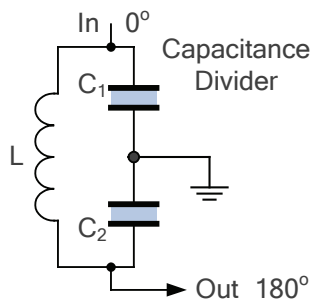
When voltage is firstly applied to the circuit, transistor  $TR_1$  starts conducting. As the collector current of  $TR_1$  increases, it is coupled through the  $0.1\mu\text{F}$  capacitor to the tank circuit, causing it to oscillate. This  $0.1\mu\text{F}$  capacitor also acts as an isolation capacitor to prevent any DC from flowing through the tank circuit. The output oscillations are fed-back through the  $0.2\mu\text{F}$  feedback capacitor. The feedback fraction in this example is 2.

Commonly, a Hartley oscillator uses one single tapped inductor instead of two separate inductors or with an FET instead of a bipolar transistor. In the past Hartley oscillators with different inductive coils were common, but less so today due to size, weight and cost.

## 4. THE COLPITTS OSCILLATOR

The **Colpitts Oscillator** configuration resembles that of the previous shunt-fed *Hartley oscillator*. The difference this time is that the tapping of the LC resonance tank circuit is now made using a capacitive voltage divider network, rather than a tapped inductor to provide the required regenerative feedback making it more stable than the Hartley.

**FIGURE 5. COLPITTS OSCILLATOR TANK CIRCUIT**



The Colpitts oscillator uses a capacitive voltage divider network as its feedback source. The two capacitors,  $C_1$  and  $C_2$  are placed across a single common inductor,  $L$  as shown in Figure 5. Then  $C_1$ ,  $C_2$  and  $L$  form the tuned tank circuit with the condition for oscillations being:  $(X_{C1} + X_{C2}) = X_L$ .

The advantage of this type of capacitive circuit configuration is that with less self and mutual inductance within the tank circuit, frequency stability of the oscillator is improved along

with a more simple design.

Since the frequency of oscillations is determined by the resonance of the tank circuit. The frequency of oscillations for the Colpitts oscillator is given as:

$$f_r = \frac{1}{2\pi\sqrt{LC_T}} \quad \text{Where: } C_T = \frac{C_1 C_2}{C_1 + C_2}$$

Clearly, the amount of feedback depends on the values of  $C_1$  and  $C_2$ . The feedback factor ( $\beta$ ) is determined by the ratio of these two capacitors which in turn, governs the excitation of the oscillator. The feedback factor or ratio is simply given as:

$$\text{Feedback Factor} = \frac{C_1}{C_2} \%$$

A Colpitts feedback factor is typically 0.1 to 0.5, or 10 to 50%. Too much feedback may cause the output to become distorted, while small amounts may prevent oscillations.

As with the Hartley oscillator, the Colpitts oscillator uses a single stage bipolar transistor amplifier as the gain element which produces a sinusoidal output.

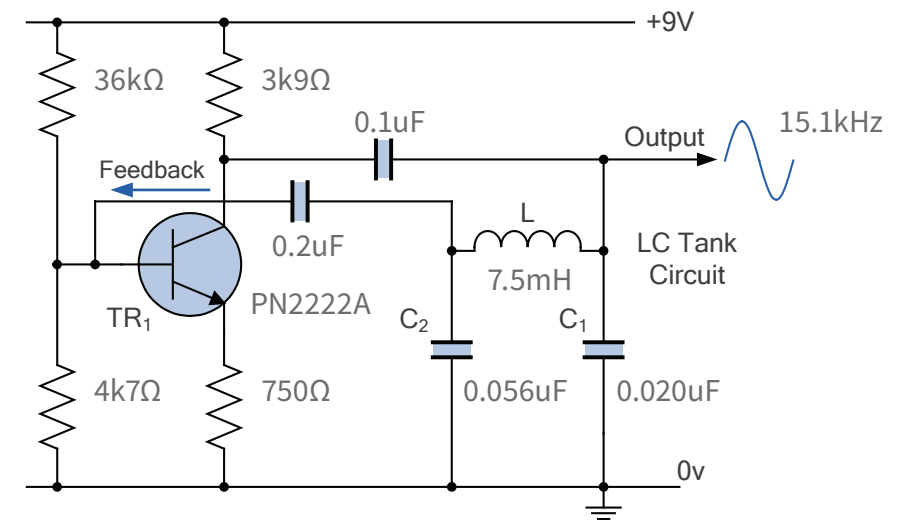
Since the oscillator is a common-emitter configuration, collector voltage is  $180^\circ$  out-of-phase with base voltage.

*The Hartley oscillator uses two coils in its tank circuit. The Colpitts oscillator uses two capacitors in its tank circuit*

The arrangement of the capacitors in the voltage-divider network produces the desired  $180^\circ$  phase shift resulting in regenerative or positive feedback.

Thus, if we require a **15kHz** Colpitts oscillator, the inductor would be **7.5mH** with a **0.02uF** and a **0.056uF** capacitor giving a feedback factor  $\beta$  of 0.35, or 35% as shown in Figure 6.

**FIGURE 6. COLPITTS OSCILLATOR**



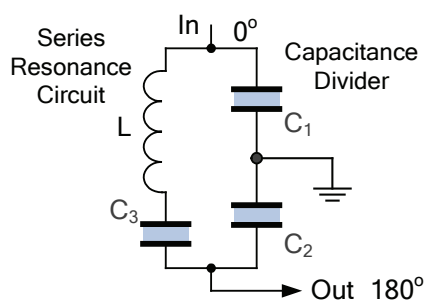
The tapped capacitor arrangement identifies it as a Colpitts oscillator. Control of the oscillatory output frequency is primarily provided by  $C_2$ . Commonly, the two capacitors are variable types and ganged together allowing for adjustment of the output frequency.

## 5. THE CLAPP OSCILLATOR

The **Clapp oscillator** is another type of LC feedback oscillator circuit and a variation of the previous Colpitts oscillator. The only difference is the addition of another capacitor in series with the inductive coil tank circuit to allow for tuning of the output frequency.

The capacitive voltage divider produces the feedback factor as before, but the addition of the new capacitor,  $C_3$  forms a series resonant circuit with the inductor allowing capacitor tuning without affecting the feedback factor which is  $C_1/C_2$ .

FIGURE 7. CLAPP OSCILLATOR TANK CIRCUIT



Thus the tank circuit consists of a series LC resonance circuit in parallel with a capacitive divider network.

The circulating tank current flows through all three capacitors  $C_1$ ,  $C_2$ , and  $C_3$  which are effectively in series. Thus the equivalent total capacitance,  $C_T$  used to calculate the resonant frequency is given as:

$$C_T = \frac{1}{1/C_1 + 1/C_2 + 1/C_3} \quad \text{Standard capacitors in series equation}$$

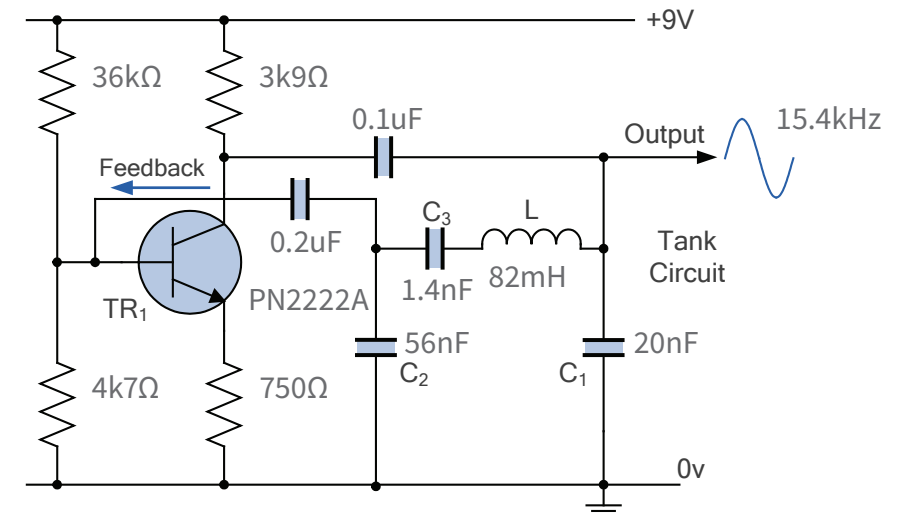
However, since capacitor  $C_3$  is generally much smaller in relation to the two tank capacitors  $C_1$  and  $C_2$ , tuning capacitor  $C_3$  effectively becomes  $C_T$ . That is:  $C_T = C_3$ , which determines the oscillation frequency ( $f_r = 1/(2\pi\sqrt{LC_3})$ ).

Using the 15kHz Colpitts oscillator circuit of Figure 6. If  $C_1 = 0.02\mu\text{F}$  (20nF) and  $C_2 = 0.056\mu\text{F}$  (56nF) Then for an inductive coil of 82mH, tuning capacitor  $C_3$  will be: 1.4nF.

$$\frac{1}{2\pi\sqrt{LC_3}} = \frac{1}{2\pi\sqrt{0.082 \times 1.3 \times 10^{-6}}} = 15.4\text{kHz}$$

Note that the capacitive value of  $C_3$  at 1.4nF is almost identical to the calculated value of 1.3nF, proving that  $C_T$  is approximately equal to  $C_3$ , for a given resonant frequency. The theory of operation of the Clapp oscillator circuit is identical to the Colpitts discussed in the previous section.

FIGURE 8. CLAPP OSCILLATOR



Then we can see that the Clapp oscillator is a variation of the Colpitts oscillator making use of series resonance, instead of parallel resonance, in the tuned circuit. Capacitors  $C_1$  and  $C_2$  determine the feedback factor which are large compared to  $C_3$ .

Thus, the sinusoidal oscillation frequency is determined by series resonance circuit of  $L$  and  $C_3$  since  $C_3$  has almost no effect on feedback.

*The Clapp oscillator is a Colpitts with a series-resonant tank circuit*

The Clapp oscillator has very good frequency stability and is unaffected by variations in load conditions. Therefore, it is commonly used in radio frequency (RF) receiver and transmitter applications.

## 6. THE RC PHASE-SHIFT OSCILLATOR

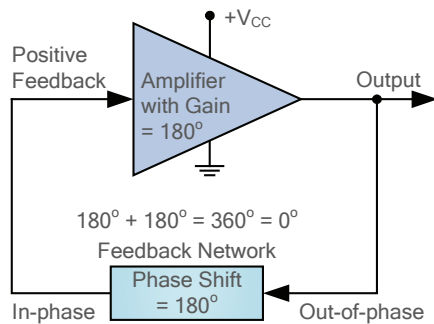
One of the disadvantages of LC (inductance-capacitance) sinusoidal oscillators is that they use wound inductors or transformers in their tank circuit to determine the oscillation frequency.

In the low and audio frequency ranges, LC oscillators are generally not practical as their inductors would need to be large and expensive. But we can replace the inductive-capacitive (LC) tank-circuit components with one that uses resistors and capacitors only.

**RC oscillators** use resistors and capacitors to determine the oscillation frequency which makes them inexpensive, easy to construct and tune. There are two basic types of RC oscillator which can be used to produce sinusoidal waveforms. The *Phase-shift oscillator* and the *Wien-bridge oscillator*.

As its name implies, the phase-shift oscillator uses a conventional amplifier and a phase shifting RC feedback network. A single stage transistor amplifier will produce  $180^\circ$  of phase shift between its output and input signals when connected as a common-emitter inverting amplifier.

**FIGURE 9. FEEDBACK CIRCUIT**



In an RC network, we can make use of the fact that a phase shift occurs between the input and the output of the same network by using interconnected RC elements.

If we use enough RC elements, the input will be shifted  $180^\circ$  through the RC network and  $180^\circ$  again through an inverting amplifier stage to produce the required positive (regenerative) feedback.

This then gives us " $180^\circ + 180^\circ = 360^\circ$ " of phase shift which is effectively the same as  $0^\circ$ , thereby giving us the required positive feedback. In other words, the total phase shift of the feedback loop should be  $0^\circ$ , or any multiple of  $360^\circ$  to obtain the same effect.

We recall that in a purely capacitive circuit, current leads the voltage by  $90^\circ$ . In an RC network the phase difference between current and voltage is at some value between  $0^\circ$  and  $90^\circ$ , because the resistance affects the phase relationship.

However resistance does not vary with frequency, but a capacitor is a frequency sensitive device since its reactance ( $X$ ) changes with frequency. Then the phase difference in an RC circuit is a function of its resistance ( $R$ ) and its capacitive-reactance ( $X_c$ ).

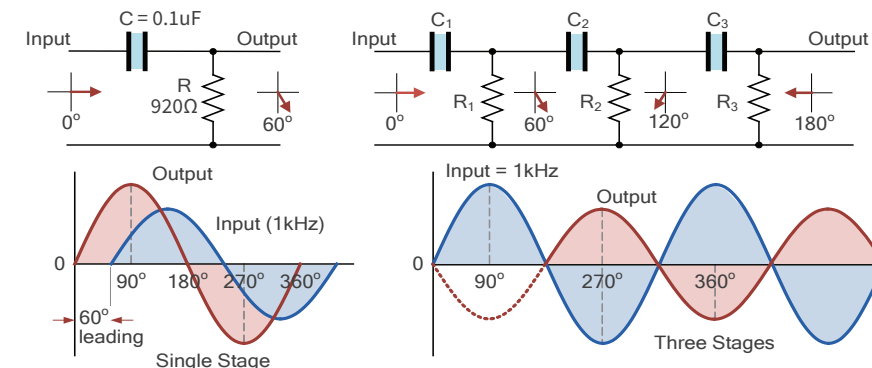
The amount of actual phase shift in the RC network depends upon the values of the resistance ( $R$ ) and the capacitance ( $C$ ), at the chosen frequency of oscillations with the phase angle ( $\phi$ ) being given as:

$$X_c = \frac{1}{2\pi fC} \quad \text{and, } R = R$$

$$\therefore \phi = \tan^{-1} \frac{X_c}{R}$$

So, if we had an RC network with a 1kHz AC input signal applied, a resistance value of **920Ω** and a capacitive value of **0.1μF** would give us the required  $60^\circ$  of phase-shift. If three such  $60^\circ$  phase-shift networks were cascaded together, as shown in Figure 10, the combined phase shift would total  $180^\circ$  since each individual RC network contributes  $60^\circ$  towards the total phase shift of  $180^\circ$  for a 1kHz input signal.

**FIGURE 10. RC PHASE-SHIFT NETWORK**



Note that while we have used here an example of three RC stages cascaded together above with the phase shift of each stage being  $60^\circ$ . It is also possible to have feedback networks with four stages ( $4 \times 45^\circ$ ) or six stages ( $6 \times 30^\circ$ ) to provide the required  $180^\circ$  of phase-shift providing increased oscillator stability.

If all the resistors (**R**) within the feedback network are of equal value, that is  $R_1 = R_2 = R_3 \dots$  etc, and the capacitors (**C**) are also equal in value,  $C_1 = C_2 = C_3 \dots$  etc. Then the frequency of oscillations produced by an RC oscillator is simply given as being:

$$f_r = \frac{1}{2\pi \sqrt{2NRC}}$$

Where:  $f_r$  is the oscillation frequency in Hertz, **R** is the resistance in Ohms, **C** is the capacitance in Farads, and **N** is the number of RC feedback stages.

So in our simple example above, the number of stages is three, so  $N = 3$  thus, ( $\sqrt{2 \times 3} = \sqrt{6}$ ). For a four stage RC feedback network,  $N = 4$  so ( $\sqrt{2 \times 4} = \sqrt{8}$ ), and so on.

Since the resistor-capacitor combination in the RC Oscillator ladder network also acts as an attenuator, it reduces the signal by some amount as it passes through each passive RC stage. We would assume that the three, phase-shift RC sections are independent of each other but no, this is not the case. The total accumulative feedback attenuation therefore becomes  $1/29^{\text{th}}$  ( $V_{\text{OUT}}/V_{\text{IN}} = \beta = -1/29$ ) across all three stages.

Then the voltage gain of the amplifying device must be sufficiently high enough to overcome these RC losses. Then in order to produce a total loop gain  $A\beta = 1$ , in our three stage RC network of Figure 10, the amplifiers voltage gain ( $A_v$ ) must be greater than **29**. Therefore resistors,  $R_f$  and **R**, of Figure 11. are related by the required gain of the amplifier.

**Operational Amplifier** phase-shift oscillators are more common and easier to configure than their bipolar transistors counterparts. The op-amp based oscillator consists of an inverting amplifier configuration producing the required  $180^\circ$  of phase-shift while the RC feedback network produces the other  $180^\circ$  at the required frequency ( $180^\circ + 180^\circ$ ). The RC phase-shift network is connected from the op-amps output back to its inverting input.

The type of feedback network where the capacitors are in series and the resistors connected to ground (0V) potential is known as a *phase-lead* configuration. In other words, the output voltage leads the input voltage producing a positive phase angle.

We can also create a *phase-lag* configuration by simply changing the positions of the RC components so that the resistors are connected in series and the capacitors are connected to ground (0V) potential resulting in the output voltage lagging the input voltage producing a negative phase angle.

Then let us use an operational amplifier as the active device to construct a 3-stage phase-lead RC phase-shift oscillator to produce a sinusoidal output frequency of 6.5kHz.

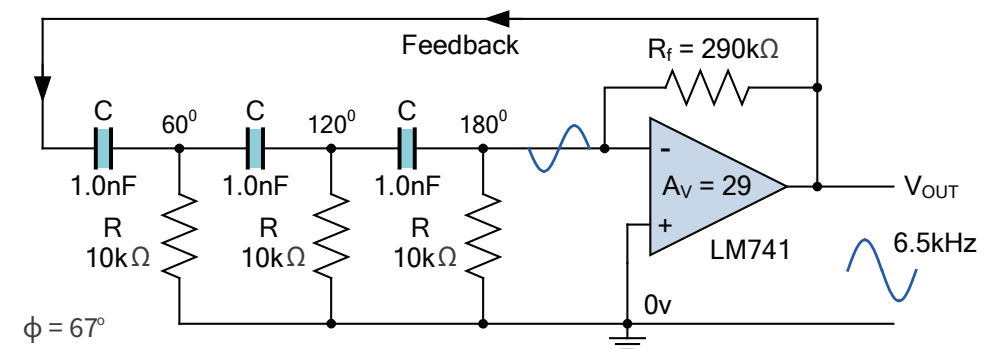
$$f_r = \frac{1}{2\pi \sqrt{2NRC}} = \frac{1}{2\pi \sqrt{6RC}} = \frac{1}{15.4RC} = 6500\text{Hz}$$

Therefore, if we use phase-shift network capacitors (**C**) of say, **1.0nF**, then the phase-shift network resistors (**R**) will be equal to: **10kΩ** as shown in Figure 11.

Then the op-amps voltage gain ( $A_v$ ) must be equal to, or slightly greater than, **29** in order to sustain oscillations. If the resistive value of the oscillators network resistors are **10kΩ**, the value of the inverting amplifiers feedback resistor  $R_f$  is calculated as being:

$$R_f = A_v \times R = 29 \times 10\text{k}\Omega = 290\text{k}\Omega$$

**FIGURE 11. 3-STAGE PHASE-LEAD RC PHASE-SHIFT OSCILLATOR CIRCUIT**



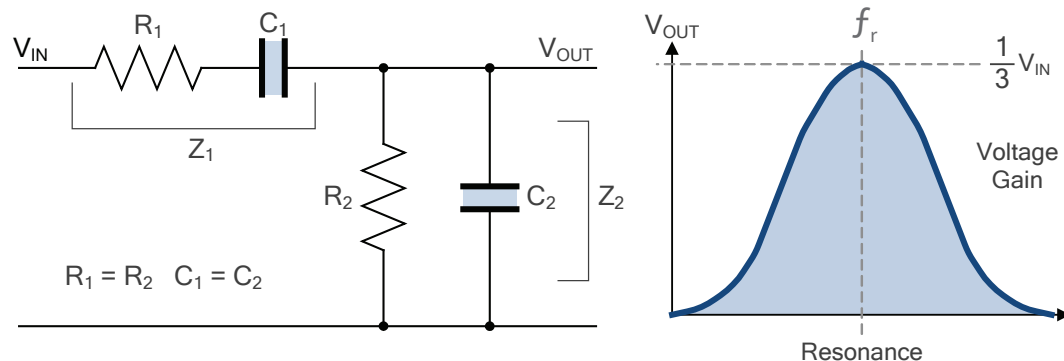
## 7. THE WIEN-BRIDGE OSCILLATOR

Like the previous phase-shift oscillator, the Wien-bridge oscillator also uses RC networks and is so called because its circuit configuration is based on a frequency-selective form of the standard Wheatstone bridge circuit.

The **Wien Bridge Oscillator** uses a feedback circuit consisting of a series and parallel combination of resistors and capacitors to form the feedback network. This produces either regenerative (positive or leading) or degenerative (negative or lagging) feedback depending upon the oscillation frequency.

Whereas before, the phase-shift RC network produce the required  $180^\circ$  of phase shift for regenerative feedback. In the Wien-bridge oscillator circuit, the RC bridge network selects the frequency at which feedback occurs, but does not shift the phase. Then at the resonant frequency  $f_r$  the oscillators phase shift is  $0^\circ$ . Consider the following lead-lag network of Figure 12.

**FIGURE 12. WIEN BRIDGE LEAD-LAG NETWORK**



At low frequencies, series capacitor  $C_1$  acts as an open circuit due to its high reactance resulting in virtually no output signal. At very high frequencies, the reactance of the parallel capacitor  $C_2$  becomes very low, shunting the output to ground, so again there is no output.

So there must be a frequency point between these two extremes of  $C_1$  being open circuited and  $C_2$  being short circuited where the output voltage,  $V_{OUT}$  reaches its maximum value, and there is. The frequency value of the input waveform at which this happens is called the oscillators *Resonant Frequency*, ( $f_r$ ). The same as for LC feedback oscillators.

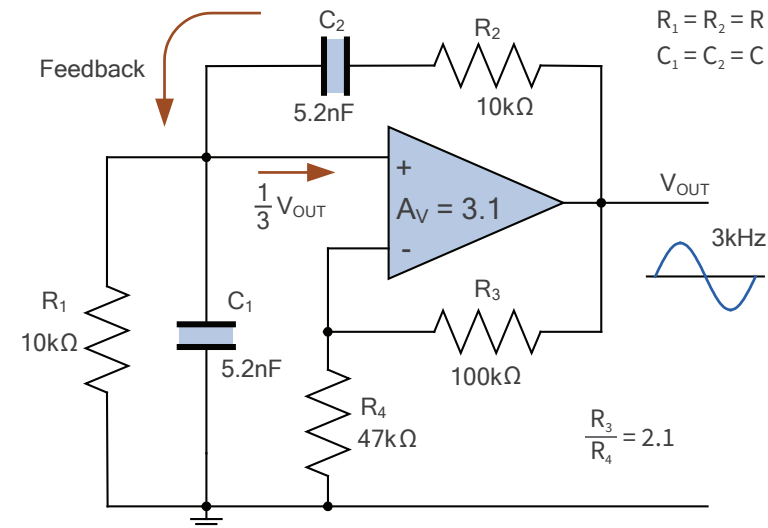
At this resonant frequency, the circuits reactance equals its resistance, that is:  $X_C = R$ , and the phase difference between the input and output reduces to zero degrees. The magnitude of the output voltage is therefore at its maximum and is equal to one about third ( $1/3$ ) of the input voltage,  $V_{IN}$ .

The generalised formula for the resonant frequency of a lead-lag network is given as:

$$f_r = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}} = \frac{1}{2\pi RC}$$

Remember, at this frequency the output of the circuit is  $1/3$  the input ( $1/3 V_{IN}$ ). If we now place this RC lead-lag network across a non-inverting operational amplifier which has a loop gain of  $1+(R_3/R_4)$  equal too or greater than three ( $A_v \geq 3$ ) a basic Wien bridge oscillator circuit is produced as shown in Figure 13.

**FIGURE 13. WIEN BRIDGE OSCILLATOR CIRCUIT**



## 8. THE TWIN-T OSCILLATOR

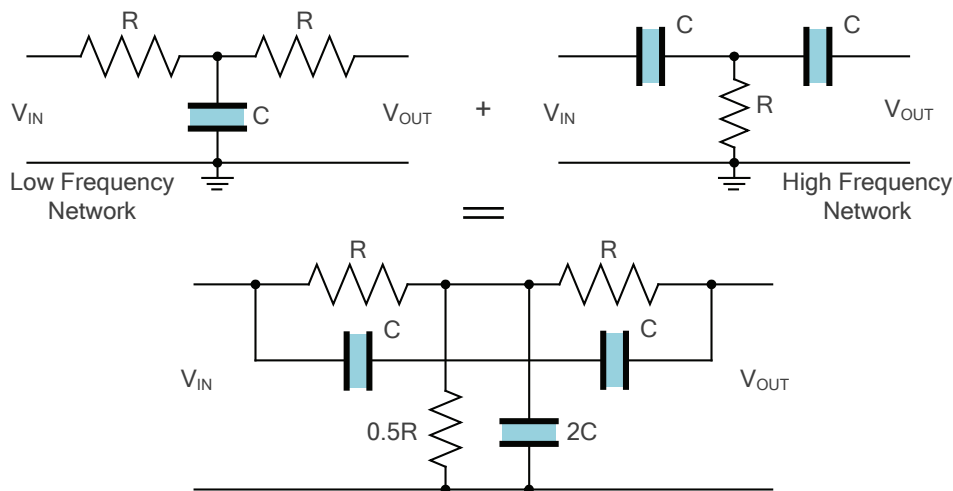
The **Twin-T Oscillator** is another type of fixed-frequency RC oscillator similar in many ways to the Wien-bridge oscillator, producing a sine wave output. The twin-T oscillator uses two “Tee” shaped RC networks in its feedback loop (hence the name).

Feedback is provided by the twin-T configured RC networks allowing some of the output signal to be fed back to the amplifier’s input terminal. One of the twin-T filter networks has a low-frequency response. While the other has a high-frequency response. The combined parallel filters produce a notched response with its center frequency equal to the desired oscillation frequency,  $f_c$ .

The twin-T RC network provides the required  $180^\circ$  of phase-shift at the given oscillation frequency, with the inverting amplifier providing the other  $180^\circ$  of phase-shift. These two conditions create  $360^\circ$  in total of phase-shift allowing for sustained oscillations.

The twin-T oscillator (sometimes known as a parallel-T oscillator) uses resistance and capacitance (RC) networks with two interconnected “T” sections (having their R and C elements in opposite formation) connected together in parallel as shown in Figure 14.

**FIGURE 14. TWIN-T RC NETWORK**



Clearly we can see in Figure 14. that one of the RC passive networks has a low-frequency response, while the other has a high-frequency response, meaning that oscillations cannot occur at frequencies above or below the tuned notch frequency due to the negative feedback path created through the twin-T network.

However, at the tuned frequency any negative feedback becomes negligible, allowing the positive feedback path created by the connected amplifying device to dominate thus creating oscillations at one single frequency.

Thus the individual twin-T networks which make up the RC network are defined by the following equations:

1. For the low-pass R-C-R network:

$$f_{R-C-R} = \frac{1}{2\pi} \sqrt{\frac{\frac{1}{C} + \frac{1}{C}}{2C \times R^2}}$$

2. For the high-pass C-R-C network:

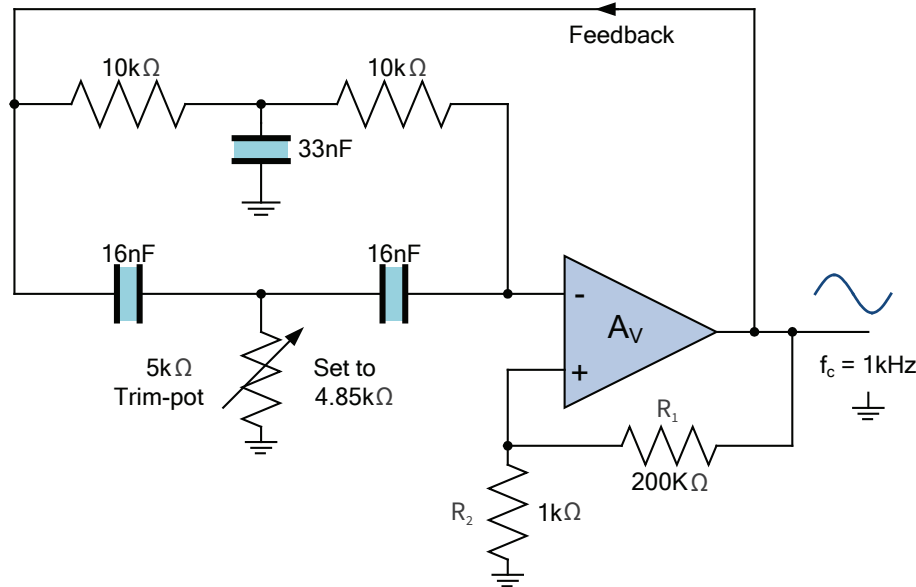
$$f_{C-R-C} = \frac{1}{2\pi} \sqrt{\frac{1}{C^2 \times 0.5R \times 2R}}$$

Combining these two sets of equations together will give us the final equation for the null or centre frequency of the notch resulting in oscillations for a twin-T network.

$$f_c = \frac{1}{2\pi RC}$$

You may have noticed that it is the same resonant frequency formula as before. Also, at the oscillating frequency,  $f_c$  the feedback gain drops to almost zero so we require an operational amplifier circuit with a loop gain much greater than one (unity). Thus we can use a non-inverting amplifier configuration shown in Figure 15. to obtain the required positive gain at the required oscillating frequency.

FIGURE 15. TWIN-T RC OSCILLATOR CIRCUIT



Then we can see that the twin-T oscillator receives its positive feedback to the non-inverting input through the voltage divider network and its negative feedback through the twin-T RC network.

Positive feedback for the op-amp required for oscillation is provided by the feedback resistor  $R_1$  while resistor  $R_2$  ensures start-up. As a general rule of thumb, for the circuit to oscillate as close to the required frequency as possible, the ratio of these two resistors needs to be greater than one-hundred ( $>100$ ). In Figure 15, the loop gain is calculated as being 201,  $(1 + (R_1/R_2))$ . Why not!

The operating frequency of Figure 15. oscillator is calculated at 1kHz, but for the Twin-T circuit to oscillate at this frequency, the fixed value “T-leg” resistor  $R/2$  can be replaced with an adjustable trimmer pot which can be adjusted to compensate for capacitor tolerances and the finite gain-bandwidth product of the op-amp.

## 9. THE RELAXATION SQUARE WAVE OSCILLATOR

Another common class of oscillator is the non-sinusoidal oscillator. As the name implies, the output from these types of oscillators is not a pure sine wave but can be rectangular, square, pulse, saw tooth, triangular, or even a combination of any such shapes to form pulse-timing and delay circuits.

These non-sinusoidal oscillators work by charging a capacitor (C) through a resistor (R), and then discharging it quickly when the voltage across the capacitor reaches some pre-determined threshold.

The oscillation frequency is determined by the time it takes for the circuit to “relax” or recover from one voltage condition to another.

*Relaxation Oscillators are made by charging a capacitor through a resistor to produce a non-sinusoidal waveform*

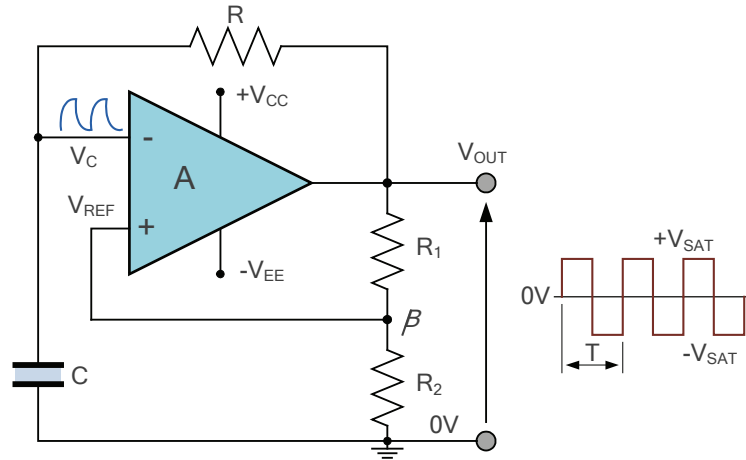
That is, during part of the oscillation cycle, energy is rapidly stored in the capacitor and is gradually released during the “relaxation” part of the cycle. Oscillators based on this principle are known as **Relaxation Oscillators**.

Both bipolar transistors (BJT's) and field effect transistors (FET's) can be used to create relaxation oscillators. The **Operational Amplifier** or *Op-amp* for short, is a very versatile device which can be used in a variety of different electronic circuits and applications including oscillator circuits.

Operational amplifiers configured as comparators can be used with positive feedback to operate as an astable or free-running relaxation oscillator to produce a fixed square or rectangular shaped output waveform.

By placing an RC timing circuit across the output of an operational amplifier, we can use the capacitors charging voltage,  $V_c$  to change the output state of the op-amp from one saturation level to another indefinitely as shown in Figure 16.

FIGURE 16. OP-AMP RELAXATION OSCILLATOR CIRCUIT



Lets assume that the capacitor,  $C$  of Figure 16 is fully discharged and the output of the op-amp is saturated at the positive supply rail,  $+V_{CC}$ . The capacitor starts to charge up from the output voltage,  $V_{OUT}$  through the feedback resistor,  $R$  at a rate determined by their RC time constant.

As soon as the capacitors voltage,  $V_c$  at the op-amps inverting (-) terminal is equal to, or greater than, the reference voltage,  $V_{REF}$  at the non-inverting terminal (the op-amps output voltage fraction divided between resistors  $R_1$  and  $R_2$ ), the output will change state and be driven to the opposing negative supply rail,  $-V_{EE}$ .

But the capacitor, which has been happily charging towards the positive supply rail ( $+V_{CC(sat)}$ ), now sees a negative voltage,  $-V_{EE(sat)}$  across its plates. This sudden reversal of the output voltage causes the capacitor to quickly discharge towards the new value of  $V_{OUT}$  at a rate dictated again by their RC time constant.

Once the op-amps inverting terminal reaches the new negative reference voltage at the non-inverting terminal, the op-amp once again changes state and the output is driven to the opposing supply rail voltage,  $+V_{CC(sat)}$ . The capacitor now see's a positive voltage across its plates and the charging cycle begins again. Thus, the capacitor is constantly charging and discharging creating a square shaped output waveform.

The timing period of the output waveform is determined by the RC time constant of the RC components and the feedback fraction,  $\beta$  established by  $R_1$  and  $R_2$ , which also sets the reference voltage  $V_{REF}$ . The expression given for the timing period of oscillations is:

$$\beta = \frac{R_2}{R_1 + R_2}$$

$$T = 2RC \times \ln \left[ \frac{1 + \beta}{1 - \beta} \right] \text{ Seconds}$$

Where:  $R$  is Resistance,  $C$  is Capacitance,  $\ln()$  is the Natural Logarithm of the feedback fraction, and  $T$  is periodic time in seconds.

Then we can see from the above equation that the frequency of oscillation, which is given as  $f = 1/T$ , not only depends upon the RC time constant but also upon the feedback fraction,  $\beta$ .

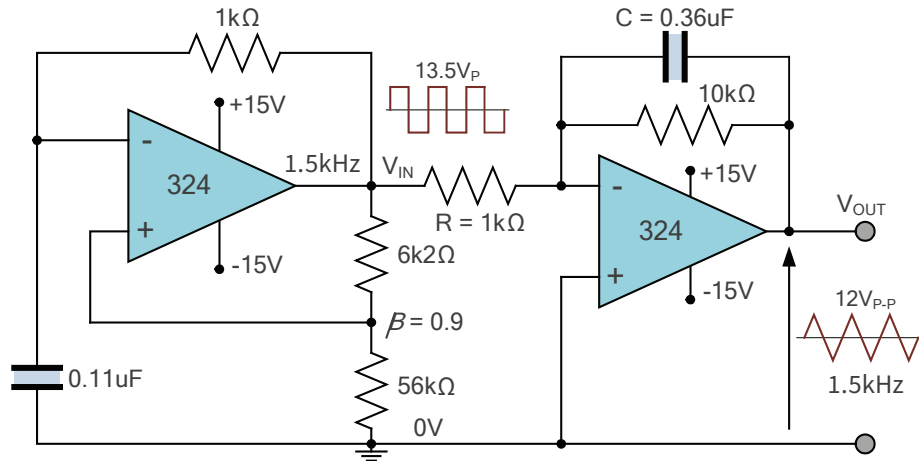
However, if we used resistor values that gave a feedback fraction of **0.462**, ( $\beta = 0.462$ ), then the frequency of oscillation of the circuit would be equal to just  $1/(2RC)$  since the linear log term becomes equal to one.

## 10. THE TRIANGULAR WAVE OSCILLATOR

We can take this basic op-amp circuit of Figure 16. one step further by cascading the relaxation oscillator with an op-amp integrator circuit. We can use the rectangular shaped output waveform of the relaxation oscillator to drive the input of an integrator to produce a triangular output waveform as shown in Figure 17.

The op-amp integrator is basically an inverting amplifier whose output voltage is proportional to the negative integral of the input voltage simulating the mathematical operation of integration. That is, its output responds to changes in the input voltage over time producing an output voltage which is proportional to the integral of the input.

FIGURE 17. OP-AMP TRIANGULAR OSCILLATOR CIRCUIT



Clearly, the triangular output waveform will have the same period and frequency as the relaxation oscillator, but we can calculate its peak-to-peak output voltage value as being:

$$V_{OUT(P-P)} = \frac{V_{IN(P)}}{2fRC}$$

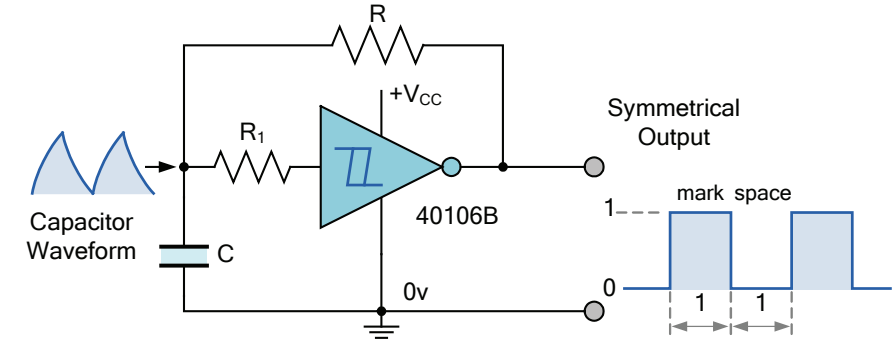
Where:  $V_{IN(P)}$  is the peak input voltage, and  $f$  is the input oscillation frequency. In Figure 17 example, this is 13.5V<sub>p</sub> and 1.5kHz respectively giving  $V_{OUT}$  as 12V<sub>P-P</sub>.

## 11. SCHMITT INVERTER RELAXATION OSCILLATOR

Finally, we can create very simple digital relaxation oscillators using a basic Schmitt trigger action inverter such as the TTL **74LS14**, or CMOS **40106** to produce square wave clock or digital timing signal.

Schmitt inverters change state between an upper and a lower threshold level as the input voltage signal increases and decreases about the input terminal. This upper threshold level “sets” the output and the lower threshold level “relaxes” the output which equates to a logic “0” and a logic “1” respectively for an inverter as shown in Figure 18.

FIGURE 18. BASIC SCHMITT INVERTER OSCILLATOR



Input resistor  $R_1$  prevents the capacitor from damaging the CMOS input circuitry as it discharges rapidly at high frequencies. The mark-space ratio is more evenly matched at about 1:1 when the feedback resistor, ( $R$ ) is below 100kΩ resulting in a smaller timing capacitor,  $C$ . The frequency of oscillation,  $f$  is given as being approximately  $1/(1.2RC)$ .

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Last revision: August 2023

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With the completion of this Oscillators eBook you should have gained a good and basic understanding and knowledge of the different types and operation of sinusoidal and non-sinusoidal oscillators. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering as well as the study of electronic feedback and relaxation oscillator circuits.

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